

Generative AI Analysis Report: A VAE for Handwritten Digits

MNIST Digit Generation and Representation Learning

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Overview

This report addresses a generative modeling task: learning to represent and generate images of handwritten digits. A Variational Autoencoder (VAE) was implemented in PyTorch and trained on the MNIST handwritten digit dataset to (1) learn a compact latent representation of digit images and (2) generate new digit images by sampling from that latent space and decoding the samples.

Dataset or Prompt Description

The data source is MNIST (LeCun, Cortes, & Burges, 1998), a collection of 70,000 real, handwritten digit images (60,000 for training, 10,000 for testing), each a 28x28 grayscale scan of a digit from 0-9. MNIST was obtained via `torchvision.datasets.MNIST`, which downloads the official archive automatically; it is publicly available, not synthetic, and has not been used in any earlier project in this capstone series. The dataset supports this generative task directly: its images are small, centered, and consistently formatted, which lets a compact model learn a meaningful latent space without first solving unrelated problems like cropping, alignment, or color normalization.

Model Design and Training Approach

The VAE consists of a fully connected encoder (784 $\rightarrow$ 400 $\rightarrow$ 400, producing a mean and log-variance for a 20-dimensional Gaussian latent distribution), a reparameterization step that samples the latent code in a differentiable way, and a symmetric fully connected decoder (20 $\rightarrow$ 400 $\rightarrow$ 400 $\rightarrow$ 784) ending in a sigmoid so outputs fall in $[0, 1]$ (Kingma & Welling, 2014). A fully connected rather than convolutional architecture was used because MNIST digits are small, single-channel, and centered, a setting in which a plain feed-forward VAE is a well-established, stable baseline. The model was trained for 15 epochs with the Adam optimizer, minimizing the negative evidence lower bound (ELBO): the sum of a binary cross-entropy reconstruction term and a closed-form KL-divergence term that regularizes the latent distribution toward a standard normal prior (Kingma & Welling, 2014). A VAE was chosen over a GAN specifically for training stability: a VAE optimizes a single differentiable objective via ordinary backpropagation, avoiding the adversarial min-max dynamics that make GAN training comparatively difficult to stabilize.

Output Evaluation and Interpretation

Three checks were used to evaluate the trained model beyond the training loss curve alone: sampling new digits directly from the prior $N(0, 1)$ and decoding them (testing pure generation), reconstructing held-out test images (testing how much information the encoder-decoder pipeline preserves), and interpolating linearly between the latent codes of two real digits (testing whether the latent space is smooth and semantically organized rather than a disconnected lookup table). Training was stable throughout: total loss decreased smoothly and monotonically from 177 to 102 per image over 15 epochs, with the reconstruction and KL terms both moving sensibly (reconstruction loss fell steadily while KL divergence rose and then plateaued, indicating the encoder was organizing the latent space rather than collapsing it). The 64 freshly generated samples covered all 10 digit classes with no mode collapse, and most were immediately recognizable as a specific digit. Reconstructions of clean, typically-written test digits were close to faithful, while reconstructions of unusually drawn digits came out blurrier and less recognizable, showing the model handles common handwriting styles better than outliers. The latent-space interpolation between a real '0' and a real '1' produced a smooth, gradual morph with no abrupt jumps, evidence of a continuous, well-organized latent space, though the interpolation's middle frames were not valid digits at all -- a reminder that only the regions of latent space the training data actually populated decode to meaningful images. The average test-set loss (102.23 per image) closely matched the final training loss (101.79), indicating the model generalized rather than memorizing training examples. The most consistent weakness across every output was blurriness: a plain VAE trained with a pixel-wise reconstruction loss tends to average over several plausible sharp outputs, producing visibly soft images rather than crisp ones (Doersch, 2016).

Ethical Considerations and Responsible Use

Generative models of handwriting, even a small MNIST-scale digit model like this one, sit on a technology continuum that includes systems capable of generating convincing forged signatures or handwritten text; the same encoder-decoder and latent-sampling approach demonstrated here, scaled up and trained on signature or full-handwriting data, is a documented category of misuse risk for document fraud and identity impersonation, so any extension of this kind of model toward realistic handwriting synthesis would need explicit safeguards and a legitimate use case. More broadly, generative AI systems raise open questions about creative ownership and consent: when a generative model produces new content after training on a corpus of real human-created work, questions of who is authoring that output and what obligations exist toward the original data's creators remain actively debated rather than settled (Epstein et al., 2023). Even though MNIST digits are simple, anonymized shapes rather than the personal or creative works usually at the center of that debate, the same underlying question applies to any generative system built on human-produced training data: responsible use requires being transparent that outputs are machine-generated and being deliberate about what data the model is allowed to be trained on and reproduce.

Limitations and Future Improvements

The current model uses a fully connected architecture, which ignores the 2D spatial structure of the images; a convolutional encoder/decoder would likely produce sharper, more spatially coherent reconstructions with a similar or smaller parameter budget. Fifteen training epochs and a fixed 20-dimensional latent space were also a specific, untuned choice; a systematic sweep over latent dimensionality and a longer training run (with early stopping based on validation loss) could improve reconstruction fidelity. The evaluation in this report is entirely qualitative aside from the one held-out loss figure; a natural next step would be a quantitative generative-quality metric such as Frechet Inception Distance, plus a classifier-based check (feeding generated digits into a pretrained digit classifier and checking whether its predicted labels look reasonable) to measure sample quality less subjectively than eyeballing a grid of images.

References

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